MATTHEW RABURN

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SUMMARY

Engineer who turns operational pain points into AI tooling that ships and gets used. Designed and built Rumil, a production RAG assistant in daily use across an enterprise support organization (over 1,000 questions a month): over a million Elasticsearch chunks across eight product brands, hybrid BM25 and vector retrieval with cross-encoder reranking, and a 17-tool agentic loop on Azure OpenAI GPT-5.1, backed by golden-set evals and around 19,700 automated tests. Eight years in enterprise support, PLM administration, and infrastructure. Builds with Python, FastAPI, TypeScript, and Claude Code.

CORE COMPETENCIES

RAG and hybrid retrieval (Elasticsearch, Azure AI Search, BM25 and dense vectors, cross-encoder reranking), agentic tool-use loops, LLM evaluation and golden-set eval harnesses, Azure OpenAI, Anthropic Claude API, Claude Code, Python, FastAPI, pytest, PowerShell, TypeScript, React, Next.js, Node.js, REST APIs, Docker and Docker Compose, GitHub Actions CI/CD, Azure (Container Apps, Entra ID, Key Vault), AWS (EC2, S3, IAM, Lambda, RDS, Bedrock), SQL Server, PostgreSQL, Snowflake, Playwright, Linux, Windows Server, Active Directory, Intune, SCCM, Office 365, Windchill PLM, C•CURE 9000 and iSTAR access control

EXPERIENCE

Johnson Controls – Westford, MA

L3 Technical Support Engineer, Security Products | Sep 2025 - Present

• Designed and built Rumil, an internal RAG assistant indexing over a million Elasticsearch chunks across eight product brands, using hybrid BM25 and vector retrieval, cross-encoder reranking, and a 17-tool agentic loop on Azure OpenAI GPT-5.1; in daily use across the support organization, with over 1,000 questions a month.

• Took Rumil from a homelab prototype to a supported platform: Azure Container Apps first, then a JCI-hosted production VM with Docker Compose, Microsoft Entra ID authentication, rate limiting and request tracing, GitHub Actions CI/CD on self-hosted runners, around 19,700 automated tests, Playwright end-to-end specs, and a golden-set eval harness; built throughout with Claude Code as the daily development harness.

• Measured it: golden-set recall@10 of 0.82; 48 of 50 correct in an unrehearsed head-to-head against the vendor assistant bundled with the case tool (26 of 50); seven guardrails cut forced retries on ungrounded answers from 25% to 0%; an embed-cache fix cut nightly tokens from 22.9M to 2.17M.

• Built AI-driven log and crash-dump analysis tooling plus PowerShell and Python diagnostics utilities, cutting manual triage time and mean time to resolution on critical site outages.

• Resolve high-priority L3 escalations for C•CURE 9000 access-control environments, including SQL replication conflicts, certificate and network failures, install and upgrade failures, and full site outages at universities, hospitals, and federal customers, partnering with software engineering to isolate bugs and ship hotfixes.

• Troubleshoot integration issues between C•CURE 9000 and iSTAR controllers, readers, and third-party security hardware; author technical advisories and Tier 3 documentation that reduce repeat escalations from L1/L2 teams.

Pall Corporation – Remote

Senior Engineering Systems Analyst | Jan 2021 - Aug 2025

• Main technical administrator for the engineering software suite (CAD, PLM, simulation) across global operations, owning stability, vendor compliance, security, and data integrity.

• Drove AI-powered automation for log analysis, repetitive tasks, and troubleshooting, saving hours weekly for engineering teams.

• Technical lead for the Pall Water divestiture: migrated 3 TB+ of technical data three weeks ahead of schedule with zero incidents.

• Managed simulation infrastructure and vendor contracts exceeding $1M, and stabilized Catia and Windchill Workgroup Manager for a $300M business unit.

• Cut workstation setup time by 72% by scripting and automating SCCM deployments of Autodesk Suite, Ansys, and Creo View across 800+ global endpoints.

• Authored and maintained the SOP library for installations, licensing, and system changes, reducing escalations and shortening onboarding.

• Administered Windchill Workgroup Manager, Catia, and SolidWorks PDM for global engineering users, including FlexLM license servers.

Boyd (joined as Lytron; acquired Jul 2019) – Woburn, MA

Systems Administrator | Oct 2019 - Jan 2021

• Primary systems administrator for a 280-person office: servers, network infrastructure, user accounts, and security.

• Built JavaScript and PostgreSQL integrations syncing PLM and ERP systems, removing manual re-entry between the two.

• Led the Office 365 and Microsoft Teams rollout for 100+ staff.

Database Administrator | Feb 2018 - Oct 2019

• Cut Engineering Change Notice cycle time by 75%, from 14 days to 3.5 days, by rebuilding the workflow and its documentation.

• Scripted validation into PLM workflows, cutting part-creation errors.